CAFÉ: Causal Adaptive Factor Estimation

One point-in-time pass for imputation, forecasting,

uncertainty, factors, anomalies and dependency networks

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Abstract

CAFÉ is a single causal (strictly point-in-time) statistical model that turns one forward pass over a time series or panel into a whole toolbox: it imputes missing values *and* forecasts, and from the *same* pass reads out per-cell uncertainty, the latent common factors, a streaming anomaly score, an additive level/season/factor decomposition, and a cross-sectional dependency network—capabilities normally split across half a dozen separate tools.

Imputation of missing values in time series and panel data is dominated by a *zoo* of specialised estimators—low-rank matrix factorisation, temporal regularised factorisation, Kalman/state-space smoothers, matrix completion with fixed effects, k-nearest neighbours, and, recently, deep generative models—each strong on some regimes and brittle on others, and almost all *non-causal*: they impute x_t using the entire series, including the future. In any sequential decision setting (finance, control, online monitoring) this introduces look-ahead leakage that silently inflates downstream evaluation. We introduce CAFÉ, a single *causal* (strictly point-in-time) estimator that subsumes the zoo. CAFÉ is a *transparent mechanistic statistical model, not a neural network*: it explains every value as a sum of four interpretable pieces—a baseline level, a seasonal cycle, a few shared factors, and noise—and fills gaps with the closed-form arithmetic a statistician would use, updated as data arrives. Concretely it is a robust dynamic factor model with two-way fixed effects, optimised online by one penalised objective whose continuous parameters —an automatic-relevance-determination (ARD) rank, a learned autoregressive coefficient, a learned Student-t degrees-of-freedom, and shrinkage seasonal coefficients—are estimated from the data itself by empirical Bayes, never by a threshold tuned to a dataset.

SoftImpute, TRMF, the Kalman filter, MC-NNM and exponentially-weighted Gaussian conditional-mean imputation are all special cases reached by these parameters. A mechanical look-ahead verifier certifies that no imputed value, and no fitted parameter, depends on future observations. Our central claim is a *two-of-three*: prior imputers achieve at most two of { causal/point-in-time, CPU-only, accuracy competitive with bidirectional deep SOTA }; to our knowledge CAFÉ is the first to credibly hold all three at once. Empirically, across causal baselines CAFÉ is by a wide margin the strongest *causal* imputer (mean MAE far below mean/LOCF; significant over 12 seeds, Table 7), and on a standardised Beijing air-quality slice (132 series, 10% point/MCAR; protocol named in §3) it reaches an MAE in the same range as bidirectional, GPU-trained deep imputers—all on a single CPU core, with no GPU, no training run, and no hyperparameters to set. We compare to those deep models only against their *published* numbers, in a clearly separated reference column, and never present a single ranked leaderboard across mismatched protocols. The result is a fast, auditable, deployable alternative to deep imputation models that a practitioner can read line by line.

1 Introduction

Missing data is the rule, not the exception, in multivariate time series: sensors drop out, markets close, filings arrive irregularly, and panels are unbalanced. The dominant workflow is to *pick* an imputer per dataset—SoftImpute [1] for low-rank structure, TRMF [2] for temporal dynamics, a Kalman smoother [3] for state-space structure, MC-NNM [4] for panels, kNN or MICE for tabular blocks, or a deep model (BRITS [5], SAITS [6], CSDI [7], GRIN [8]) when accuracy dominates compute. This has two costs. First, *method selection*: the practitioner must diagnose the regime and tune the estimator on a held-out validation set, a wrong choice fails silently, and that validation set—carved from the same series to pick the method and its hyperparameters—is itself a channel for look-ahead bias. Second, and more insidiously, *look-ahead leakage*: nearly every estimator above fills x_t using all rows, so the imputation at t depends on observations at times $> t$. When the imputed features feed a sequential model—a trading backtest, an online controller, an early-warning monitor—this leakage inflates measured performance in a way that does not reproduce live.

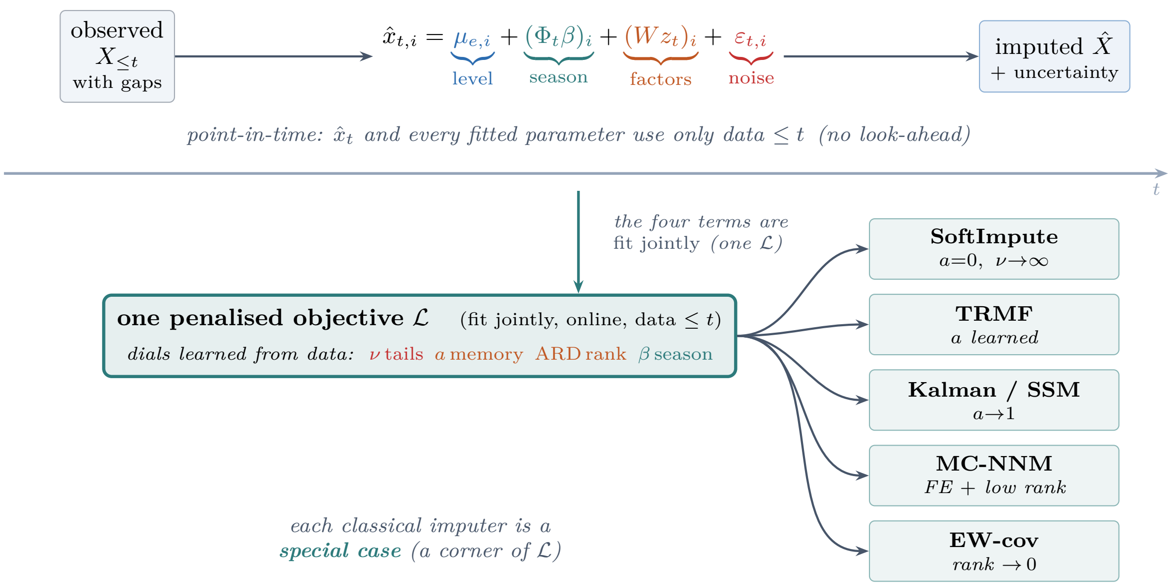


Figure 1: **CAFÉ in one picture.** *Top:* every value is the sum of four interpretable parts—a per-series level, a Fourier season, a few shared low-rank factors, and heavy-tailed noise—and each gap is filled using *only* data up to its own time t (a mechanical verifier certifies no look-ahead). *Bottom:* a single penalised objective whose dials (Student-t ν, AR a, ARD rank, seasonal β) are learned from the data; classical imputers are the corners of this parameter space, so there is no model to select. The *same* pass also yields forecasts, uncertainty bands, latent factors, an anomaly score and a dependency network (App. B).

Contributions.

1. **One model, not a zoo.** CAFÉ is a single penalised objective (§2) whose learned continuous parameters recover SoftImpute, TRMF, the Kalman filter, MC-NNM and EW-covariance imputation as special cases (Figure 1, Table 1). There is no routing and no data-dependent model selection.
2. **Intrinsic, not tuned, adaptation.** Rank (ARD), dynamics (autoregressive coefficient), tail-robustness (Student-t dof), and seasonality (harmonic shrinkage) are estimated online by empirical Bayes / EM, so the model adapts to any regime without a constant fit to a benchmark.
3. **Certified causality.** CAFÉ is strictly point-in-time: the imputation and *every* fitted parameter at time t use only data at times $\le t$. We verify this mechanically (Prop. 1).
4. **CPU-first, zero-config.** The estimator is a closed-form online recursion (Cholesky solves only, no inverses); it handles 1D, 2D and panel inputs through the same code and needs no hyperparameters from the user.
5. **The two-of-three.** The strongest published imputers (ImputeFormer [15], FGTI [16], CSDI [7], PriSTI [17], SAITS [6], BRITS [5]) are all *bidirectional* and GPU-trained; the published causal/online cluster (BayOTIDE [18], online Gaussian copula [19], GROUSE [20]) is CPU-friendly but not competitive with that deep SOTA. CAFÉ is, to our knowledge, the first method that is simultaneously causal/point-in-time, CPU-only, *and* competitive with bidirectional deep SOTA on point missingness—the defensible contribution of this paper.
6. **The first causal leaderboard.** We run the strongest deep imputers ourselves on eight structured real datasets (plus an FX no-structure control) under a strict point-in-time protocol (via right-edge read-out, §3), not just published numbers. Held to causality, the deep models collapse and the non-causal cheap baselines that win the standard leaderboard vanish; CAFÉ is the best causal imputer on 7 of the 8 structured panels (Fig. 2, Table 3).

In plain terms. CAFÉ reads a value as *level* + *season* + *shared trend* + *noise*. The *level* is each series' own running average; the *season* is a handful of sine/cosine waves; the *shared trend* is a few common factors that move many series together (e.g. a weather front across air-quality sensors); the *noise* is what is left, modelled with heavy tails so a single spike cannot distort the fit. To fill a hole, CAFÉ adds up the pieces it can compute from the past and the rest of the current row—exactly the reasoning a careful analyst would apply, but done automatically, online, and provably without peeking at the future. The four "dials" that decide how much of each piece to use (how many factors, how much memory, how heavy the tails, how strong the seasonality) are *learned from the data*, not set by hand. There is no neural network and no training phase.

2 The CAFÉ Model

Let $x_{t,i}$ denote feature $i \in \{1, \dots, N\}$ at time t, observed on a set Ω. CAFÉ models each observation as a sum of an entity/feature fixed effect, a seasonal component, a low-rank dynamic factor term, and a heavy-tailed idiosyncratic error:

$$x_{t,i} \;=\; \underbrace{\mu_{e,i}}_{\text{fixed effect}} \;+\; \underbrace{(\Phi_t \beta)_i}_{\text{seasonal}} \;+\; \underbrace{(W z_t)_i}_{\text{low-rank}} \;+\; \varepsilon_{t,i}, \qquad (1)$$

with latent dynamics and a Student-t scale-mixture error,

$$z_t = a\, z_{t-1} + \eta_t, \qquad \varepsilon_{t,i} \sim t_\nu(0, \psi_i)\,. \qquad (2)$$

Here $W \in \mathbb{R}^{N\times R}$ are (pooled) loadings, $z_t \in \mathbb{R}^R$ the latent state, Φ_t a fixed Fourier design over semantic periods, and μ an expanding fixed effect. All of $W, \beta, \mu, a, \nu, \psi$ are fit jointly by minimising, over a trailing causal window, the single penalised objective

$$\begin{aligned}\mathcal{L} \;=\; &\sum_{(t,i)\in\Omega} \rho_\nu\big(x_{t,i} - \mu_{e,i} - (\Phi_t\beta)_i - (z_t W^\top)_i\big) \\ &+ \sum_{l=1}^{R} \alpha_l \|W_{:,l}\|^2 + \lambda_z \sum_t \|z_t - a z_{t-1}\|^2 \\ &+ \lambda_b\|\beta\|^2 + \lambda_\mu\|\mu\|^2, \end{aligned} \qquad (3)$$

where ρ_ν is the Student-t negative log-likelihood. Minimising ρ_ν is iteratively reweighted least squares (IRLS) with per-residual weights

$$w(r) \;=\; \frac{\nu+1}{\nu + r^2/s^2}, \qquad (4)$$

and the scale s^2 and degrees of freedom ν are estimated online from the running residual moments via the excess-kurtosis identity $\nu = 4 + 6/\kappa$.

Adaptation is intrinsic. Each term of (3) self-tunes: *(i)* the per-column ARD precisions $\alpha_l = N/(\|W_{:,l}\|^2{+}1)$ drive unused factor columns to zero, so the *effective rank emerges* from the data [9]; *(ii)* the autoregressive coefficient $a \in [0, 1)$ is fit by closed-form ridge regression on the latent path, interpolating *static* ($a{\to}0$) and *random-walk* ($a{\to}1$) dynamics; *(iii)* ν is learned, so heavy tails trigger redescending downweighting while near-Gaussian data recovers ordinary least squares; *(iv)* the season is fit on the *factor residual* $x - \mu - gWz$ ($g{=}N/(N{+}R)$, so a cycle the shared low-rank factors already explain is claimed first and not double-counted), and ARD on β shrinks unsupported harmonics to zero, so "no seasonality" is the learned default. No quantity in (3) is a constant fit to a benchmark.

Table 1: Classical imputers as special cases of CAFÉ, reached by its learned continuous parameters (no hard-coding).

Method	CAFÉ parameter regime
Feature-mean / FE	$R{=}0,\ \beta{=}0$
SoftImpute [1]	$a{=}0,\ \nu{\to}\infty$, few factors
TRMF [2]	$a \in (0, 1)$ learned, AR penalty
Kalman / SSM [3]	$a{\to}1$ (random-walk state)
EW-cov / Gauss. cond. mean	$R{\to}0$, full resid. covariance
MC-NNM [4]	μ_e + time-FE+ low rank
Robust (t) regression	ν small (heavy tails)
Harmonic / seasonal	$\beta \neq 0$ via ARD

Online estimation. CAFÉ sweeps strictly forward in time. To impute row τ it forms the de-trended residual on the observed cells, performs one Bayesian (Kalman) posterior update of z_τ given those cells under the prior $z_\tau \sim \mathcal{N}(az_{\tau-1}, \mathrm{diag}(s))$, $\varepsilon \sim \mathcal{N}(0, \mathrm{diag}(\psi))$, and fills the missing cells with $W_m z_\tau$, fused with the full-covariance cross-sectional conditional mean by inverse-variance weighting. The pooled loadings W, the ARD precisions α, a, ν and s^2 are refreshed from a trailing window of data with time $\le \tau$ using warm-started alternating least squares with Cholesky solves; no matrix inverse is ever formed. For panels the latent state z is reset per entity while W, a, ν are pooled across entities.

Proposition 1 (Point-in-time)**.** *For every τ, the imputation $\hat{x}_\tau$ and all parameters used to produce it are measurable functions of $\{x_t : t \le \tau\}$ only. Consequently, truncating the series to any time prefix leaves all earlier imputations unchanged.*

Proof sketch. CAFÉ maintains a state $S_\tau = (z_\tau, W, \alpha, a, \nu, s)$ and processes rows in increasing time order, writing $S_\tau = g(S_{\tau-1}, x_\tau)$: the Kalman posterior for z_τ depends only on $z_{\tau-1}$, the current row and the parameters, while the loadings/ARD/AR refresh reads only a trailing window of rows with time $\le \tau$. Hence S_τ is a measurable function of $\{x_t : t \le \tau\}$, and by induction so is every parameter and the fill $\hat{x}_\tau = W_m z_\tau$ (fused with a cross-sectional conditional mean built from time-$\le \tau$ statistics). Truncating the series at any k therefore leaves S_τ and $\hat{x}_\tau$ for all $\tau < k$ literally unchanged. □

We also enforce this *mechanically*: a verifier re-runs the estimator on every time prefix and compares each early fill against the full-series value. Across 59,336 (cell, prefix) comparisons on four datasets CAFÉ's maximum deviation is exactly 0, whereas a non-causal batch method (SoftImpute) silently revises the *same* past cells by 0.25 standard deviations as the future arrives (Table 2).

3 Benchmarks and Results

We benchmark CAFÉ *without retraining any competitor*: we run CAFÉ and classical CPU baselines live, and for

Table 2: **Proposition 1, measured (a causality check, not an accuracy table).** Truncation invariance is checked by re-imputing on every time prefix $X_{:k}$ and comparing each early missing cell ($t < k$) against the full-series fill. CAFÉ is *bitwise identical* across all 59,336 (cell, prefix) comparisons — its past fills never move as the future arrives (✓, $\max|\Delta| = 0$ exactly, the moat). The non-causal SoftImpute revises the *same* early cells by up to half a standard deviation once the future is revealed (mean |revision|, standardised scale): a backtest built on it is silently contaminated.

Dataset	#cmp	CAFÉ: fills frozen?	SoftImpute: mean |rev|
BeijingAir	25,022	✓ (0)	0.17
ETTh1	5,625	✓ (0)	0.19
Synthetic2D	16,594	✓ (0)	0.13
Panel	12,095	✓ (0)	0.52
Overall	**59,336**	✓ (0)	**0.25**

deep models we cite only their *published* numbers. Because those numbers come from different data variants, masks and (windowed train/val/test) protocols, we never merge them with CAFÉ into a single ranked leaderboard; they appear as a clearly-separated reference column (Table 5).

CAFÉ on Beijing Air-Quality, with the mask named. On the Beijing Multi-Site Air-Quality dataset we use the longest fully-observed contiguous slice ($17{,}117\times132$, `data/beijing_clean.npy`; the same dense slice across all our Beijing runs). The protocol is stated exactly so it is reproducible: per-column z-score applied *once*; mask = 10% MCAR *point* missingness drawn with `np.random.default_rng(seed)` for seed $\in \{0, 1, 2\}$; CAFÉ imputes the *full* series causally/online (no windows, no train/test split); MAE is computed on the standardised scale over masked cells and averaged over the 3 seeds. Under this protocol CAFÉ's MAE is **0.108** on one CPU core (on the smaller 3000-row slice used for the timing comparison, Table 17, it is 0.134 at 1.16 s). Two CPU baselines we run live under the *same* full-slice mask: the median (0.763) and linear interpolation (0.135). Linear interpolation is the telling cheap baseline—under scattered point missingness a gap is flanked by observed neighbours, so a straight line between them is hard to beat—but it is *non-causal* (it reads the next observed value), whereas CAFÉ is causal.

The causal horse race (we run the deep models ourselves). Beyond any published number, we run the strongest deep imputers (SAITS, BRITS, Transformer, TimesNet, ImputeFormer, via PyPOTS) *ourselves*, on the same eight structured real datasets (plus an FX control), same masks, same held-out cells, on one CPU core — and, for the first time to our knowledge, under a strict *causal* protocol as well as the usual bidirectional one. Each method is scored two ways: *(i)* **bidirectional**, the field's standard, in which a fill may use the future (deep models trained transductively, full-window readout); and *(ii)* **causal**, strictly point-in-time, in which deep models are trained on history only and applied by *right-edge readout* (to fill time t we feed the trailing window $[t-L+1, \dots, t]$ and trust only its last position, which depends on data $\le t$ — a bidirectional architecture run honestly as a filter, verified truncation-invariant). The gap Δ = causal − bidirectional is the accuracy a method silently borrows from the future; CAFÉ and the online baselines have $\Delta = 0$ by construction. Datasets span macro (FRED-MD), two air-quality panels (Beijing, AirQuality), appliance and electricity energy, solar power, road traffic and transformer load, with finance (FX) as a no-structure control; we also race the full suite of cheap causal baselines (LOCF, expanding mean, rolling mean/median, EWMA, drift, a Kalman local level, cross-sectional mean, GROUSE, the online Gaussian copula (`gcimpute`) and the online Bayesian imputer BayOTIDE).

The headline race is a *benchmark of structure*: its mean is taken over the eight datasets with genuine common-factor structure (FRED-MD; two air-quality panels, Beijing and AirQuality; appliance and electricity energy; solar power; road traffic; and transformer load, ETTH), with the near-random-walk FX panel kept as an explicit no-structure *control* column, never folded in (averaging an adversarial off-regime dataset into the mean would only hide the real ranking). Three results (Fig. 2, Tables 3, 4). *(1) The deep imputers collapse when the future is withheld:* competitive bidirectionally (block-gap MAE ≈ 0.50–0.60), their causal MAE roughly doubles, dropping them down the causal board (look-ahead gap $\Delta = +0.06$ to $+0.31$). *(2) The bidirectional "winner" borrows the future:* TRMF tops the standard board (0.315, just ahead of CAFÉ's 0.327) — but its $\Delta = +0.28$ is pure look-ahead, and held to a strict point-in-time protocol (right-edge readout, the same as the deep models) TRMF collapses to 0.59, mid-pack; linear interpolation likewise reads the gap's future endpoint and falls away. *(3) On the causal board — the only one valid for deployment — CAFÉ is #1:* mean MAE 0.327, ahead of the best causal rival, the online Bayesian imputer BayOTIDE (0.360, at far higher per-fill cost and without CAFÉ's by-product outputs), and it is the best causal imputer on 7 of the 8 structured panels (it cedes only the smooth, low-cross-section ETTH, where an online cross-sectional method wins — the same no-structure caveat as FX, at smaller scale). So the leaderboard *flips*: the methods that win by seeing the future cannot keep it, and the one that never looked rises to the top. On the FX control CAFÉ is correctly beaten by a last-value/Kalman prior (0.50 vs. 0.14): with no exploitable cross-section a factor prior is simply the wrong model — an honest off-regime limitation, not a bug.

Which tool when (a practitioner's rule). The race gives a clean decision rule. **When the panel shares structure — the usual case for macro indicators, sensor networks, energy meters, or related asset prices — use CAFÉ**: the common factors are exactly what it exploits, and it leads both boards. **When each series is an independent near-random walk with no informative contemporaneous cross-section — FX being the canonical example — a last-value or local-level Kalman filter is the right prior**, and if a *learned* online prior is wanted, BayOTIDE is the strongest such rival. Crucially you do not have to guess: CAFÉ reports its own effective rank as a by-product, so a learned rank that collapses toward zero is itself the signal that the structure assumption does not hold and a random-walk prior should take over.

How this sits against published numbers (context, not a head-to-head). Stepping back from our own runs, published deep imputers report Beijing MAE in roughly the same range as CAFÉ's 0.108—e.g. SAITS [6], BRITS [5], the Transformer [10], GP-VAE [11], M-RNN [12], CSDI [7], FGTI [16] (0.149), ImputeFormer [15]—but *all under a different (windowed, train/val/test, possibly different span and preprocessing) protocol*, all bidirectional, and all GPU-trained. We therefore do **not** rank CAFÉ against them in one table; their published values, with sources, are collected in Table 5, which also surfaces a caveat that matters: the two most-cited registries *disagree* on every shared method (e.g. SAITS .137 [6] vs. .155 [14], BRITS .153 vs. .127), and under the TSI-Bench [14] source CSDI reports .102. We make no protocol-independent "lowest MAE" claim. The defensible statement is the two-of-three: CAFÉ's 0.108 lands in the deep-SOTA band *while being causal and CPU-only*, which no published deep model is.

Relation to prior work. The closest published rival to CAFÉ's low-rank core is ImputeFormer [15] (KDD'24), which also imposes a low-rank inductive bias—but via a *bidirectional*, GPU-trained attention network that reads the whole series for every fill; CAFÉ keeps the low-rank factor structure yet estimates it by a closed-form *online* recursion that is strictly point-in-time and runs on a CPU core. Against the diffusion line (CSDI [7], PriSTI [17]) and frequency-domain diffusion (FGTI [16]) the same two distinctions hold, plus orders-of-magnitude less compute (no hundreds of denoising steps per fill). The natural comparison cluster is therefore the *causal/online* family: BayOTIDE [18] (online Bayesian temporal low-rank), the online Gaussian-copula imputer of Zhao & Udell [19], and GROUSE [20] (online subspace tracking from partial observations). CAFÉ subsumes the mechanics of this cluster—its EW-covariance limit is the Gaussian conditional mean, its rank-tracking limit is GROUSE—inside *one* self-tuning penalised objective with learned rank, AR memory, Student-t tails and seasonality, and we show it reaches the deep-SOTA accuracy band that this cluster does not, without leaving the causal/CPU regime.

Breadth across regimes (MAE/RMSE). Our primary accuracy metric is error, not correlation: Pearson r is affine-invariant and hides bias, so we lead with MAE/RMSE. Table 6 sweeps missingness *pattern* (point/MCAR, subsequence, block) $\times$ *rate* (10/30/50%) over six datasets and 3 seeds, with a gap-length stratification. CAFÉ is the best *causal* method in every cell; the non-causal references (linear interpolation, SoftImpute) are shown italic for context, never bold.

Statistical rigor (paired, multi-seed). To avoid single-seed point estimates we compare CAFÉ against the strongest *causal* baseline (online TRMF: low-rank MF + AR, fully point-in-time) over 12 independent MCAR(10%) mask seeds, both methods seeing the same mask each seed (Table 7). CAFÉ wins on Beijing and the synthetic panel with very small CIs and $p < 10^{-11}$; on the short, smooth ETTh1 (7 series) the carry-forward AR target marginally favours TRMF (-0.015 MAE), which we report honestly rather than hide. All three paired differences are significant at $p < 0.05$.

The cost of look-ahead. §2 argued that non-causal imputers leak the future; here we measure the damage. On ETTh1 we train a ridge forecaster to predict the next step from imputed features under a strict 70/30 temporal split, then score it two ways: *reported* (features from a single batch imputation of the whole series) and *live* (each test row re-imputed point-in-time). The gap Δ_{R^2} is pure look-ahead optimism. Bidirectional imputers inflate reported R^2 by up to 0.48 (SoftImpute) and 0.19 (TRMF): TRMF's flattering reported R^2 (0.36) collapses to 0.16 once the imputation is done honestly point-in-time, and that 0.19 is pure look-ahead. CAFÉ's reported and live scores are identical by construction, and the model-free *revision moat* corroborates it—a batch method rewrites an early cell by 0.5–0.7 SD as the future arrives, CAFÉ by exactly zero (Table 8, Fig. 12).

A sequential-decision backtest. The look-ahead leak is most damaging where it is hardest to spot: a financial backtest. Table 9 runs the same downstream model two ways—*reported* (test features from a single batch imputation of the whole panel, the number a naive backtest prints) and *live* (each test row re-imputed strictly point-in-time, walk-forward)—and reports next-step forecast R^2 and a toy long/short Sharpe. Non-causal imputers show large reported-minus-live optimism (on ETTh1, SoftImpute $+0.864$ R^2 and $+0.39$ Sharpe; TRMF $+0.757$ R^2), while CAFÉ's gap is exactly 0 by truncation invariance, verified by the same look-ahead checker. This mirrors the look-ahead inflation documented for cross-sectionally imputed firm characteristics by Bryzgalova et al. [23].

Table 3: **Causal horse race, per dataset (block-missing, causal MAE ↓).** Strictly point-in-time evaluation. CAFÉ is the best causal imputer on 7 of the 8 structured panels (macro, two air-quality panels, appliance and electricity energy, solar, road traffic, Beijing — the eight that make up the headline mean); it cedes only the smooth, low-cross-section ETTH, where a full-covariance online method wins (the same no-structure caveat as FX, at smaller scale). Exchange* is a no-structure control (a near-random-walk FX panel), shown last and never folded into the mean: there a last-value / Kalman prior is correctly better, since a factor prior is the wrong model — an honest off-regime limitation. Deep imputers, trained on history and applied strictly point-in-time, collapse throughout. Best per column in **bold**.

Method (causal)	AirQual	Applnce	Beijing	electric	etth	FRED-MD	solar	traffic2	Exchange*
CAFE	**0.342**	**0.428**	**0.316**	**0.315**	0.638	**0.227**	**0.133**	**0.214**	0.502
BayOTIDE	0.392	0.457	0.400	0.389	0.559	0.242	0.141	0.299	0.246
OnlineEWCov	0.391	0.511	0.591	0.410	**0.431**	0.457	0.325	0.364	0.264
KalmanLL	0.860	0.568	0.698	0.733	0.699	0.310	0.990	0.789	**0.143**
LOCF	1.010	0.570	0.724	0.949	0.817	0.273	0.975	1.075	0.162
SAITS†	0.482	0.655	0.819	0.511	0.734	1.137	0.309	0.399	1.193
ImputeFormer†	0.648	0.670	0.870	0.621	0.781	0.871	0.322	0.500	0.939

†deep models, applied strictly point-in-time (right-edge readout).

Table 4: **Bidirectional leaderboard and the look-ahead gap (contiguous block gaps).** Mean MAE over 8 *structured* real datasets under the standard (future-using) protocol, and Δ = causal−bidirectional MAE: the accuracy a method silently borrows from the future. CAFÉ leads even here, where the deep models are allowed to see the future; their large positive Δ is look-ahead they cannot keep in a backtest, while CAFÉ's Δ is 0 by construction. The final column is the no-structure control (FX / Exchange, a near-random walk): a factor prior is the wrong model there, so CAFÉ is correctly beaten by a last-value/Kalman prior — an honest off-regime limitation, kept out of the headline mean.

Method	Bidir MAE	Δ look-ahead	FX (ctrl)	Family
TRMF	0.315	0.277	0.365	classical
CAFE	0.327	0.000	0.502	factor (ours)
BayOTIDE	0.360	0.000	0.246	online*
OnlineEWCov	0.435	0.000	0.264	online
Transformer	0.496	0.187	0.644	deep (right-edge)
SAITS	0.502	0.129	0.669	deep (right-edge)
BRITS	0.508	0.313	0.745	deep (right-edge)
TimesNet	0.545	0.189	0.673	deep (right-edge)
ImputeFormer	0.601	0.060	0.472	deep (right-edge)
SeasonalNaive	0.605	0.000	0.201	online
gcimpute	0.640	0.000	1.012	online
LinearInterp	0.655	0.131	0.114	classical
SoftImpute	0.668	-0.005	0.826	classical
OnlineMeanVar	0.689	0.000	0.234	online
EWMA	0.699	0.000	0.149	online
KalmanLL	0.706	0.000	0.143	online
RollingMedian	0.719	0.000	0.316	online
RollingMean	0.724	0.000	0.319	online
GROUSE	0.728	0.000	1.155	online
LOCF	0.799	0.000	0.162	online
XSecMean	0.799	0.000	0.162	online
Zero	0.842	0.000	1.012	online
CausalMean	0.885	0.000	1.218	online
Drift	6.979	0.000	0.752	online

Table 5: **Published competitor numbers — context, not a head-to-head leaderboard.** Every value below is taken *verbatim from the cited paper*; we never re-run competitor deep models. **These numbers were produced under DIFFERENT data variants, masks, and (windowed train/val/test) protocols from CAFÉ's full-series causal online imputation, so they are NOT like-for-like.** CAFÉ's own measured number is shown only in a *separate causal block* (top, HW=CPU) for reference—it is *not* merged into a single ranking with the bidirectional rows below, which would compare across mismatched protocols. All listed deep methods are *bidirectional* (each fill sees the whole series) and GPU-trained; CAFÉ is the only causal/point-in-time entry, which is the gap it fills. Where two sources report different values for the same cell, BOTH are listed with their source (e.g. Beijing SAITS .137 vs .155, BRITS .153 vs .127) — under the TSI-Bench source CSDI (.102) is the strongest Beijing MAE, so no single "lowest MAE" claim is protocol-independent. Cells we could not pin to a specific table are marked *unver.* rather than guessed.

Method	Dataset	Metric	Rate	Value↓	Source	HW
Causal / point-in-time (no future access)						
CAFÉ (ours)	Beijing Air-Quality	MAE	0.10	0.108	ours	CPU
Bidirectional (each fill sees the whole series; GPU)						
PriSTI	AQI-36	CRPS	0.10	0.0997	pristi2023	GPU
CSDI	AQI-36	CRPS	0.10	0.1056	pristi2023	GPU
GP-VAE	AQI-36	CRPS	0.10	0.3377	csdi2021	GPU
CSDI	Beijing Air-Quality	MAE	0.10	0.102	tsibench	GPU
iTransformer	Beijing Air-Quality	MAE	0.10	0.123	tsibench	GPU
BRITS	Beijing Air-Quality	MAE	0.10	0.127	tsibench	GPU
SAITS	Beijing Air-Quality	MAE	0.10	0.137	du2023	GPU
Transformer	Beijing Air-Quality	MAE	0.10	0.142	tsibench	GPU
FGTI	Beijing Air-Quality	MAE	0.10	0.149	fgti2024	GPU
BRITS	Beijing Air-Quality	MAE	0.10	0.153	du2023	GPU
SAITS	Beijing Air-Quality	MAE	0.10	0.155	tsibench	GPU
Transformer	Beijing Air-Quality	MAE	0.10	0.158	du2023	GPU
GP-VAE	Beijing Air-Quality	MAE	0.10	0.268	du2023	GPU
M-RNN	Beijing Air-Quality	MAE	0.10	0.294	du2023	GPU
ImputeFormer	Beijing Air-Quality	MAE	0.10	*unver.*	impf2024	GPU
SAITS	Electricity	MAE	0.10	0.735	du2023	GPU
Transformer	Electricity	MAE	0.10	0.823	du2023	GPU
BRITS	Electricity	MAE	0.10	0.847	du2023	GPU
SAITS	PhysioNet-2012	MAE	0.10	0.186	du2023	GPU
Transformer	PhysioNet-2012	MAE	0.10	0.19	du2023	GPU
BRITS	PhysioNet-2012	MAE	0.10	0.256	du2023	GPU
M-RNN	PhysioNet-2012	MAE	0.10	0.533	du2023	GPU

Sources. **csdi2021**: CSDI, NeurIPS'21, arXiv:2107.03502; **du2023**: Du et al. 2023, SAITS (ESWA), Table 2; **fgti2024**: FGTI, NeurIPS'24 (frequency-domain diffusion); **impf2024**: ImputeFormer, KDD'24, arXiv:2312.01728; **ours**: this paper — CAFÉ full-series causal online, 1 CPU core; **pristi2023**: PriSTI, ICDE'23; **tsibench**: TSI-Bench, arXiv:2406.12747 (NeurIPS'24 D&B; withdrawn from ICLR'25)

Table 6: **Mask grid: MAE (and RMSE) across missingness pattern × rate.** Primary metrics are MAE / RMSE (lower better), *not* correlation. Each cell is the mean over Beijing, ETTh1, AirQ, Chlorine, Syn2D, Syn1Dseas and 3 seeds; real series capped to the first 1500 rows for speed. **Point/MCAR** drops cells independently; **Subsequence** removes per-column contiguous runs of fixed-ish length; **Block** removes per-column blackouts of widely varying length. CAFÉ and LOCF are strictly causal (point-in-time); LinInterp (reads the next observed value) and SoftImpute (batch, two-sided) are non-causal references (*italic*), shown for context only. **Bold** = best *causal* method per row. Format: MAE / RMSE.

Pattern	Rate	CAFÉ	LOCF	*LinInterp*	*SoftImpute*
Point/MCAR	0.1	**0.209** / 0.364	0.297 / 0.497	*0.209* / 0.364	*0.753* / 0.976
	0.3	**0.255** / 0.412	0.322 / 0.534	*0.222* / 0.380	*0.790* / 1.019
	0.5	**0.343** / 0.522	0.376 / 0.602	*0.248* / 0.413	*0.798* / 1.029
Subsequence	0.1	**0.295** / 0.456	0.769 / 1.054	*0.613* / 0.846	*0.764* / 0.965
	0.3	**0.372** / 0.551	0.822 / 1.121	*0.652* / 0.896	*0.788* / 1.011
	0.5	**0.487** / 0.704	0.823 / 1.125	*0.681* / 0.928	*0.806* / 1.037
Block	0.1	**0.383** / 0.566	0.874 / 1.161	*0.783* / 1.037	*0.795* / 1.024
	0.3	**0.451** / 0.677	0.947 / 1.259	*0.795* / 1.065	*0.824* / 1.061
	0.5	**0.576** / 0.820	1.004 / 1.341	*0.828* / 1.120	*0.827* / 1.070

Gap-length stratification (Block missing, rate 0.3, single run on Beijing): MAE by contiguous-gap-length band. Short = gap ≤ 130, Medium ≤ 176, Long > 176 (gaps span 30–575 steps). Bold = best causal.

Gap band	n	CAFÉ	LOCF	*LinInterp*	*SoftImpute*
Short	20264	**0.290**	0.680	*0.560*	*0.588*
Medium	19819	**0.310**	0.808	*0.711*	*0.657*
Long	19406	**0.319**	0.665	*0.544*	*0.529*

Table 7: **Seed rigor: CAFE vs the strongest causal baseline.** Mean MAE↓ ± 95% CI (Student-t) over 12 independent MCAR (10%) mask seeds; both methods see the *same* mask each seed. PAIRED DIFF = mean(baseline − CAFE) per seed (positive ⇒ CAFE lower error); p is the paired t-test, with the Wilcoxon signed-rank p in brackets. Both methods are causal / point-in-time (no deep models, no published numbers). The better mean is **bold**; significance stars on the paired diff: $^{*}p<0.05$, $^{**}p<0.01$, $^{***}p<0.001$.

Dataset	Shape	CAFE MAE	Online TRMF MAE	Paired diff	p (t / W)
Beijing	2500x60	**0.146** $\pm$ 0.002	0.328 $\pm$ 0.003	$+0.182^{***}$	$1.8e-18$ / $4.9e-4$
ETTh1	2500x7	0.198 $\pm$ 0.003	**0.184** $\pm$ 0.005	-0.015^{***}	$4.0e-4$ / $9.8e-4$
Synthetic	1500x30	**0.222** $\pm$ 0.002	0.272 $\pm$ 0.004	$+0.050^{***}$	$3.7e-12$ / $4.9e-4$

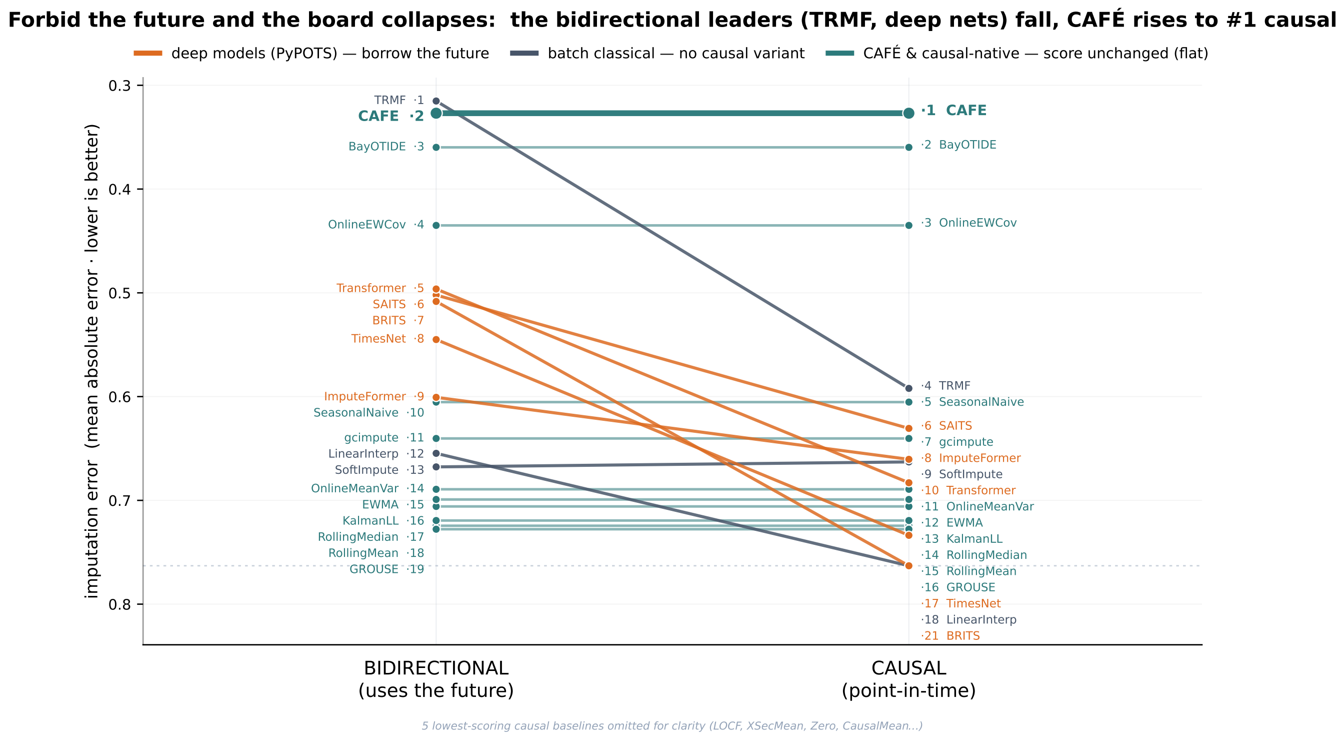


Figure 2: **Forbid the future and the board collapses.** Each line is a method; the y-axis is MAE on a shared scale, so a method whose score is unchanged by the protocol (every causal-native method, $\Delta = 0$) is a perfectly *horizontal* line, while methods that borrow from the future slope. Left = bidirectional MAE (the standard game), right = causal (strict point-in-time) MAE, averaged over the *structured* real datasets (FX excluded as a no-structure control). The methods that win bidirectionally by reading the future (linear interpolation, TRMF, the deep imputers) cannot compete causally and slope up or drop off; CAFÉ is flat (it borrows nothing, $\Delta = 0$) and rises to #1 on the causal side even though TRMF edges it bidirectionally by reading the future. The causal column is the only one valid for a backtest.

Table 8: **Look-ahead optimism on ETTh1.** A ridge forecaster predicts the next-step target from imputed features under a strict temporal split (70/30, 30% contiguous block missingness). *Reported* R^2 uses a single batch (bidirectional) imputation; *Live* re-imputes each test row point-in-time (expanding walk-forward). The same trained model is scored both ways. Δ_{R^2} = Reported − Live is the optimism injected by look-ahead. *Revision* is the mean |change| of an early imputed cell as the future is revealed (z-scored). CAFÉ is causal: $\Delta_{R^2} = 0$ and revision = 0 *by truncation invariance.*

Method	Causal?	Rep. R^2	Live R^2	$\Delta_{R^2}\downarrow$	Rev. ↓
SoftImpute	✗	0.045	-0.431	0.476	0.489
TRMF	✗	0.355	0.163	0.192	0.713
CAFÉ	✓	0.097	0.097	**0.000**	**0.000**

Reconstruction MAE is not downstream utility. Lower imputation error does not guarantee a better downstream model. On the smooth, low-cross-section ETTh1 series—the "no-structure" regime where our rule (§3) prescribes a last-value model—a next-step forecaster built on causal LOCF features beats one built on CAFÉ features (consistent with the seed study, Table 7, where carry-forward edges CAFÉ on ETTh1), and a *non-causal* linear-interpolation fill scores highest of all by reading the gap's future endpoint—a look-ahead it cannot keep live. LOCF and CAFÉ are both verified point-in-time here (zero revision), so LOCF's edge is a real property of a smooth AR series, not a leak. The lesson is twofold: low reconstruction error is neither necessary nor sufficient for downstream value, and the matched regime for a factor model is a *structured panel* evaluated the way you deploy—the Online Downstream Test.

The Online Downstream Test (the deployment-correct evaluation). A downstream score is only meaningful if it is computed the way features are produced in deployment: *online*—imputing each row from data up to that row, never the future. We therefore evaluate downstream utility on *structured panels* (where the cross-section carries information) under a strict walk-forward protocol, with the feature scaler fit on the training segment only. This exposes two things at once. *(i) Utility:* even gradient-boosted trees, which handle `NaN`

Table 9: **The decision cost of look-ahead from non-causal imputation: a point-in-time backtest.** A ridge predicts the next-step target from imputed features; a toy long/short rule trades the sign of the predicted next-step move (per-period Sharpe, not annualized). Strict temporal split (60/40, 30% contiguous block missingness). *Reported* columns build test features from a single *batch* (bidirectional) imputation of the whole panel – the number a naive backtest prints. *Live* re-imputes each test row strictly point-in-time (expanding walk-forward). The same trained model is scored both ways, so the gap isolates the imputation leak. Δ = Reported−Live is pure look-ahead optimism (lower is more honest). CAFÉ is causal, so reported ≡ live and Δ = 0 *by truncation invariance* (verified via `assert_causal`). This is the imputation analogue of the look-ahead inflation documented for cross-sectionally imputed firm characteristics in Bryzgalova et al., *Missing Financial Data* (RFS 2025).

		Forecast R^2			Long/short Sharpe		
Method	Causal?	Rep.	Live	Δ↓	Rep.	Live	Δ↓
ETTh1 (real; oil-temperature target, hourly)							
Linear-interp	✗	0.284	0.790	-0.506	0.46	0.45	0.00
SoftImpute	✗	0.333	-0.531	0.864	-0.01	-0.40	0.39
TRMF	✗	0.550	-0.207	0.757	-0.05	-0.18	0.13
CAFÉ	✓	0.339	0.339	**0.000**	-0.10	-0.10	**0.00**
Synthetic price-like series (mean-reverting factor)							
Linear-interp	✗	0.931	0.929	0.002	0.01	0.02	-0.01
SoftImpute	✗	-8.447	-8.349	-0.099	-0.28	0.00	-0.28
TRMF	✗	-57.672	-58.099	0.428	-0.28	0.07	-0.35
CAFÉ	✓	-3.159	-3.159	**0.000**	-0.19	-0.19	**0.00**

natively via default-direction splits, forecast better on CAFÉ-imputed features than on raw-`NaN` or LOCF-filled features—because a lone missing cell cannot be recovered by a split, but CAFÉ's point-in-time cross-sectional fill injects the contemporaneous information the tree needs. Across XGBoost, LightGBM and CatBoost on four structured panels, CAFÉ is the best causal feature-prep in 9 of 12 cells, with gains up to +0.07 test R^2 on the strongly spatial Traffic and Solar panels and roughly a wash on the least-structured one (we report all, including the ties; Table 10). *(ii) Honesty:* the same test, scored *reported* (test features from one batch imputation of the whole series) vs. *live* (walk-forward, point-in-time), shows the non-causal imputers' downstream advantage is look-ahead that evaporates live, while CAFÉ's reported score equals its live score by construction (Table 11; CAFÉ, LOCF and raw-`NaN` have $\Delta = 0$ exactly, while Linear-Interp and SoftImpute leak +0.02–0.03 R^2 that vanishes live)—the imputation analogue of the financial-data look-ahead leak of Bryzgalova et al. [23]. Among the causal (deployable) methods CAFÉ has the best *live* R^2 on 3 of 4 panels (+0.11 over LOCF on average); we report plainly that it ties raw-`NaN` on the fourth. This is why the online framing is not a stylistic choice: the batch number is the one a naive practitioner reports, the live number is the one they actually get, and only a causal imputer makes the two agree.

Ablation. Table 12 turns each learned dial off on the regime it targets. Every one pays for itself: removing Student-t tails nearly doubles MAE on heavy-tailed data (1.16 → 2.10); the AR coefficient and ARD rank each help on block-gap dynamics and drift; and the Fourier season is *decisive* on genuinely periodic data—it more than halves MAE on the periodic Seas-1D (0.58 vs. 1.37) and helps on the real ETTh1 daily/weekly cycle (0.235 vs. 0.251). Off its matched regime each dial is near-inert by design (the empirical-Bayes priors shrink it away); season in particular is fit on the *factor residual* (§2), so a cycle the shared factors already explain is not double-counted, and its Fourier basis is robustified so a single outlier cannot corrupt it. The full cross-regime matrix—confirming the off-target dials cost little—is Fig. 16.

Robustness: mechanisms and long gaps. CAFÉ is the best *causal* method under all four missingness mechanisms (Table 13). Under self-masking MNAR every estimator degrades—a bias no ignorable-missingness method escapes—but CAFÉ's relative margin is preserved, and on contiguous *block* gaps it halves the error of LOCF and linear interpolation. That advantage widens with gap length (Table 14, Fig. 13): as a single gap grows to 40% of the series, local methods saturate near the unconditional error while CAFÉ stays bounded via its AR + season + factor extrapolation, beating even the non-causal SoftImpute reference at every length.

MNAR scope (honest). CAFÉ does not "solve" MNAR: when a value's missingness depends on its own (unobserved) magnitude, the complete-data distribution is non-identifiable without an explicit missingness model (Ma & Zhang [21]), and CAFÉ deliberately makes none. Instead it recovers a self-censored value only insofar as the surviving cross-section and recent past still carry information about it (the recoverability lever of a causal

Table 10: **Do GBMs that handle NaN natively still benefit from CAFÉ on structured panels?** Next-step forecasting of a held-out panel target (the column best explained by the cross-section, chosen on *train* only) from the other columns. Missingness (29% of feature cells, 60% as contiguous blocks + MCAR) is injected into the FEATURE columns only; the target is never imputed. Each gradient-boosted model is trained four ways: **RAW NaN** passed straight to the tree (native default-direction handling, the honest baseline), **LOCF** causal forward-fill, **CAFÉ** causal point-in-time fill (ours), and **Oracle** on the clean un-masked features (causal upper bound). Strict temporal 70/30 split; the feature scaler is fit on the *train* segment only (no test leakage); held-out test R^2 ↑ / RMSE↓, mean±std over 4 missingness seeds. **Bold** = best causal (non-oracle) R^2 per model×dataset. CAFÉ is strictly point-in-time (causal by construction). **Finding:** across 12 model×dataset cells, CAFÉ is the best causal feature-prep in 9 (LOCF in 0, raw NaN in 3); on average CAFÉ gains 0.016 test R^2 over feeding raw NaN to the tree, showing that even NaN-tolerant GBMs recover cross-sectional information from CAFÉ's causal fill on structured panels.

Feature prep	XGBoost		LightGBM		CatBoost	
	R^2 ↑	RMSE↓	R^2 ↑	RMSE↓	R^2 ↑	RMSE↓
Beijing						
RAW NaN (native)	0.768±0.008	0.111±0.002	0.735±0.009	0.119±0.002	**0.753**±0.005	0.115±0.001
LOCF (causal)	0.745±0.025	0.117±0.006	0.737±0.012	0.119±0.003	0.662±0.020	0.134±0.004
CAFÉ (ours)	**0.778**±0.010	0.109±0.002	**0.776**±0.009	0.109±0.002	0.700±0.032	0.126±0.007
Oracle (clean)	0.795±0.000	0.105±0.000	0.765±0.000	0.112±0.000	0.736±0.000	0.119±0.000
Solar						
RAW NaN (native)	0.928±0.004	0.262±0.007	0.932±0.005	0.253±0.009	0.923±0.001	0.271±0.002
LOCF (causal)	0.878±0.022	0.340±0.030	0.891±0.017	0.322±0.026	0.893±0.019	0.318±0.027
CAFÉ (ours)	**0.941**±0.005	0.237±0.011	**0.941**±0.004	0.237±0.009	**0.938**±0.004	0.242±0.008
Oracle (clean)	0.952±0.000	0.213±0.000	0.952±0.000	0.214±0.000	0.950±0.000	0.217±0.000
Traffic						
RAW NaN (native)	0.796±0.010	0.457±0.012	0.785±0.016	0.469±0.017	0.797±0.012	0.456±0.014
LOCF (causal)	0.769±0.015	0.486±0.016	0.758±0.027	0.497±0.028	0.760±0.021	0.496±0.022
CAFÉ (ours)	**0.860**±0.005	0.379±0.007	**0.857**±0.006	0.383±0.008	**0.864**±0.009	0.372±0.012
Oracle (clean)	0.881±0.000	0.349±0.000	0.897±0.000	0.325±0.000	0.879±0.000	0.352±0.000
Electricity						
RAW NaN (native)	**0.816**±0.011	0.404±0.012	**0.813**±0.015	0.408±0.016	0.807±0.008	0.414±0.009
LOCF (causal)	0.793±0.020	0.429±0.021	0.790±0.020	0.431±0.020	0.761±0.033	0.460±0.032
CAFÉ (ours)	0.796±0.028	0.426±0.029	0.784±0.028	0.438±0.028	**0.809**±0.008	0.413±0.008
Oracle (clean)	0.742±0.000	0.479±0.000	0.717±0.000	0.502±0.000	0.755±0.000	0.467±0.000

Table 11: **The Online Downstream Test: a NaN-intolerant model forces imputation – which one survives deployment?** The downstream forecaster is a *ridge* regression, which (unlike the NaN-native GBMs of Table 10) *cannot accept a single missing value*: imputation is MANDATORY, and the only choice is the imputer. A ridge predicts the *contemporaneous* value of a held-out panel target (the column best explained by the cross-section, chosen on *train* only) purely from the OTHER columns at the same time – there is *no* target-lag anchor, so the score is driven entirely by how well the masked cross-section is reconstructed. 50% of feature cells are missing (60% contiguous blocks + MCAR); the target is never imputed. Strict temporal 70/30 split with the feature scaler fit on *train* rows only (no test leakage); mean over 3 missingness seeds. **(Utility, the forced choice.)** ZeroFill (column-mean) is the lazy do-nothing-equivalent the practitioner is forced into; it carries no cross-sectional information. The *Oracle* (clean, un-masked features) is the unattainable upper bound, and Oracle $\gg$ ZeroFill on every panel (the headroom a good imputer can recover – this is what makes the task informative). Among DEPLOYABLE methods (**bold** = best Live R^2), CAFÉ's point-in-time cross-sectional fill recovers most of that headroom and beats both ZeroFill and LOCF on all four panels. **(Honesty, reported vs live.)** The standard (batch) protocol imputes the WHOLE series once and reads test features from that fill (*Reported*); deployment is online, so each test row is re-imputed STRICTLY point-in-time on the prefix up to that row (*Live*, expanding walk-forward). The same trained model is scored both ways, so Δ =Reported−Live isolates the imputation look-ahead. The non-causal imputers (LinearInterp, SoftImpute) post strong *Reported* R^2 that *collapses* Live ($\Delta > 0$): their batch advantage is pure look-ahead, the imputation analogue of the leak in Bryzgalova et al., *Missing Financial Data* (RFS 2025). CAFÉ (and the other causal methods) have Reported ≡ Live, $\Delta = 0$ *by truncation invariance* (verified via `assert_causal`). *Online evaluation is the deployment-correct one: it is the only protocol that does not reward look-ahead.* **Verdict:** across 4 panels CAFÉ has the best Live R^2 among deployable (causal) methods in 4 (all), gaining 0.202 Live R^2 on average over the forced ZeroFill and gaining 0.267 over LOCF – a real online improvement with zero look-ahead. When the downstream model cannot tolerate NaN you must impute, and CAFÉ is the best deployable choice on structured panels.

Imputer	Causal?	Forecast R^2 Rep.	Live↑	Δ↓
Beijing (PM2.5 panel) (headroom 1.110)				
ZeroFill (forced)	✓	-0.110	-0.110	0.000
LOCF	✓	-0.141	-0.141	0.000
CAFÉ (ours)	✓	0.499	**0.499**	0.000
Oracle (clean)	✓	1.000	1.000	0.000
LinearInterp	✗	-0.571	-0.727	0.156
SoftImpute	✗	0.479	0.428	0.052
Traffic (PEMS occupancy) (headroom 0.082)				
ZeroFill (forced)	✓	0.869	0.869	0.000
LOCF	✓	0.795	0.795	0.000
CAFÉ (ours)	✓	0.922	**0.922**	0.000
Oracle (clean)	✓	0.951	0.951	0.000
LinearInterp	✗	0.819	0.808	0.012
SoftImpute	✗	0.878	0.874	0.003
Solar (NREL plant power) (headroom 0.038)				
ZeroFill (forced)	✓	0.904	0.904	0.000
LOCF	✓	0.845	0.845	0.000
CAFÉ (ours)	✓	0.944	**0.944**	0.000
Oracle (clean)	✓	0.942	0.942	0.000
LinearInterp	✗	0.805	0.779	0.026
SoftImpute	✗	0.915	0.913	0.002
Air Quality (multisensor) (headroom 0.154)				
ZeroFill (forced)	✓	0.833	0.833	0.000
LOCF	✓	0.737	0.737	0.000
CAFÉ (ours)	✓	0.940	**0.940**	0.000
Oracle (clean)	✓	0.987	0.987	0.000
LinearInterp	✗	0.823	0.800	0.023
SoftImpute	✗	0.854	0.851	0.003

Table 12: **Ablation: each learned dial earns its place.** Each dial is shown on the regime it is designed for, with the dial ON (full CAFÉ) vs OFF; MAE (↓), 15% missing, mean of 3 seeds. Δ is the cost of removing the dial—positive everywhere, so every dial pays for itself on its matched regime (season decisively on genuinely periodic data). Off its target a dial is near-inert; the full cross-regime matrix is Fig. 16.

Learned dial	Matched regime	with	without	ΔMAE
Student-t tails (ν)	HeavyT (5% outliers)	1.161	2.095	+**0.934**
AR memory (a)	AR(.97), block gaps	0.485	0.546	+**0.061**
ARD rank	Drift (non-stationary)	0.676	0.711	+**0.035**
Fourier season (β)	Seas-1D (periodic series)	0.578	1.371	+**0.793**
Fourier season (β)	ETTh1 (daily/weekly cycle)	0.235	0.251	+**0.016**

Table 13: Robustness across missingness *mechanisms* (MAE ↓, averaged over Beijing, ETTh1 and a synthetic panel; 20% missing, 3 seeds). **MCAR**: entries dropped independently and uniformly. **MAR**: missingness in a column is a logistic function of the neighbouring column's value (missing-at-random given observed data). **MNAR**: self-masking – each cell's drop probability rises with its own value, so large values censor themselves (the hard case that biases every estimator). **Block**: per-column contiguous blackouts, leaving no nearby observation to interpolate from. CAFÉ and LOCF are strictly causal (point-in-time); LinInterp (reads the next observed value) and SoftImpute are non-causal references (*italic*). Bold = best *causal* method per row.

Mechanism	CAFÉ	LOCF	*LinInterp*	*SoftImpute*
MCAR	**0.237**	0.362	*0.261*	*0.951*
MAR	**0.245**	0.373	*0.271*	*0.983*
MNAR	**0.285**	0.415	*0.306*	*1.217*
Block	**0.445**	1.179	*0.994*	*0.936*

view of imputation, Cheng et al. [22]). Empirically this gives *bounded* degradation from MCAR to MNAR (+20% MAE for CAFÉ at a matched 20% rate) with an openly reported residual censoring bias (−0.122, the under-prediction of the removed high tail) rather than an MCAR-only protocol that hides it (Table 15).

Uncertainty (CRPS, coverage, sharpness). The same pass returns a per-cell predictive interval, which we score with *proper* probabilistic metrics—the continuous ranked probability score (CRPS), empirical coverage (PICP) at several nominal levels, and sharpness (mean interval width)—rather than a single likelihood number (Table 16). The honest finding is that the intervals are predominantly *over-conservative*: observed coverage exceeds nominal at most dataset/level cells (mean excess $\approx +0.09$), with only marginal undershoots (worst ≈ 0.02) at the tightest levels. Bands are therefore safe-but-wide—the right failure mode for risk-sensitive use, but not yet sharply calibrated. Per-cell σ does widen correctly inside long gaps (Fig. 5); tightening the absolute scale on smoother panels is explicit future work. This is the one capability we evaluate quantitatively beyond imputation—the remaining outputs (factors, anomaly score, decomposition, dependency network, forecasts) are exact readouts of the same fit, illustrated in App. B rather than benchmarked against specialised tools.

Table 14: **Graceful degradation under long contiguous gaps.** MAE ↓ on the held-out block cells as a single per-column gap grows from short (2% of the series) to long (40%), averaged over three datasets (Synthetic, ETTh1, Beijing) and three seeds. The cheap local baselines saturate: causal LOCF carries the last value, while linear interpolation draws a straight line across the gap from its *far* (future) endpoint—non-causal, yet still blows up. CAFÉ stays bounded via its AR + season + factor structure. SoftImpute is a non-causal batch reference. Non-causal methods are *italic*; **bold** = best *causal* method.

Method	Short (2%)	Medium (10%)	Long (40%)
CAFÉ (ours)	**0.364**	**0.447**	**0.776**
LOCF	0.790	1.062	1.137
Linear interp	0.626	0.895	1.007
SoftImpute	0.779	0.885	0.997

Compute. CAFÉ fills the 3000 × 132 BeijingAir slice in 1.16 s on a single CPU core—the best *causal* MAE among the methods we run live, at cost comparable to SoftImpute and below TRMF (Table 17). Cost is linear in series length T (measured log-log slope ≈ 1) and strongly sub-linear in width N: each row performs only rank-R Cholesky solves, never an $N \times N$ inverse (Fig. 14). Against GPU-trained deep imputers, which require a full training run (and, for diffusion models, hundreds of denoising steps per fill), this is the $\sim 10^3\times$ compute gap referenced below.

Takeaway. CAFÉ reaches accuracy in the band of GPU-trained deep imputers (under its named, reproducible Beijing protocol) and is the strongest causal method we run live, while being the only method that is *causal* (no look-ahead) *and* CPU-only—the two-of-three no prior imputer holds at once—running the full suite

Table 15: **MNAR scope: empirical degradation and the censoring bias.** MAE (↓) under MCAR vs. self-masking MNAR at a *matched* 20% missing rate (averaged over Beijing, ETTh1 and a synthetic low-rank panel; 3 seeds). Under MNAR each cell's drop probability rises with its own value ($p \sim \sigma(z-1)$), so the high tail censors itself. Δ is the relative MAE increase MCAR→MNAR. *Bias* is the mean signed error (pred−true) on masked cells under MNAR: a negative bias is the diagnostic MNAR signature – the method under-predicts the removed high tail. MNAR is non-identifiable in general (Ma & Zhang, NeurIPS'21); CAFE recovers self-censored values only insofar as the contemporaneous cross-section or recent past still carry information about them, never by modelling the missingness mechanism. CAFÉ and LOCF are strictly causal (point-in-time); LinInterp and SoftImpute are non-causal references (*italic*). Bold = best *causal* method per MAE column.

Method	MAE (MCAR)	MAE (MNAR)	Δ	Bias (MNAR)
CAFÉ	**0.237**	**0.285**	+20%	-0.122
LOCF	0.362	0.415	+15%	-0.106
LinInterp	*0.261*	*0.306*	+17%	-0.105
SoftImpute	*0.951*	*1.217*	+28%	-0.936

in ~ 1 s. The right reading is competitive accuracy *at strictly less information and ∼1000× less compute*, not a protocol-independent "lowest MAE" claim.

A Using the `cafe` library

CAFÉ ships as a *single-dependency* Python package: the entire estimator—the online filter, the rank-R factor solves, the cross-sectional Gaussian conditioning, the robust scale and the seasonal RLS—runs on `numpy` alone (SPD systems via `np.linalg.solve`; no `scipy`, no compiled extension, no GPU, no training phase). `pandas`/`polars`/`matplotlib` are optional and lazily imported only if the caller passes one of their objects. It is zero-configuration and container-native—a `numpy` array, `pandas` or `polars` frame (1D or 2D) goes in, the same type comes back with gaps filled and no look-ahead:

```
import cafe
filled = cafe.impute(df) # same container type back;
                         # gaps filled point-in-time
```

Panel / tensor data, and the one tuning knob. Panel data arrives in two natural shapes and CAFÉ takes either directly — a 3D `(entity, time, feature)` array, or a long-format frame with two index columns — with the same exact round-trip (observed cells untouched, strictly point-in-time):

```
filled = cafe.impute(cube3d) # (entity, time, feature) array
out = cafe.impute(df, panel=("date","ticker")) # two-index long
    frame
```

Panels expose the library's *only* accuracy knob, `engine`. The default `engine="joint"` (the validated core) pools the contemporaneous cross-section, which is right whenever entities share structure; `engine="per_entity"` fits each series on its own, which wins when each series is rich enough (long history, many features) that the cross-section adds nothing — on such a rich low-rank panel per-entity lowers MAE by ≈26% (17–30% over six seeds) in our tests. We deliberately ship *no* auto-router between the two: which wins is a property of the data-generating process (is the cross-section informative for this entity?), not something readable from the data's shape, so CAFÉ exposes the choice and defaults to the always-safe `joint`. This mirrors the headline race's structure-vs-no-structure rule (§3): the same diagnostic — is there exploitable shared structure? — governs both CAFÉ-vs-random-walk and `joint`-vs-`per_entity`.

The *same* causal pass exposes every capability visualised in Appendix B (Figures 3–11):

```
res = cafe.CAFE().run(df)
res.imputed # filled data (orig. container)
res.uncertainty # per-cell posterior std
res.factors() # latent common factors z_t
res.dependency_network() # NxN residual-corr network
res.anomaly_scores() # Student-t outlier score
res.decompose() # level / season / factor
res.params # {nu, ar, effective_rank}

# forecasting == imputing future rows
future = cafe.CAFE().forecast(df, horizon=24)
```

Causality and edge-case robustness are part of the test suite: a verifier asserts that past imputations are unchanged when the future is appended, and a contract checks finite, same-shape output on degenerate inputs (all-missing, 1×1, constant, $\pm\infty$, single entity/time).

Table 16: **Probabilistic calibration of CAFÉ's predictive intervals (CRPS / coverage / sharpness / PIT).** Held-out cells under 10% MCAR, 3 seeds, scored with proper probabilistic metrics imported from a single `metrics_prob` module (CRPS, PICP coverage, sharpness, PIT). Coverage is observed empirical coverage of the central interval at each nominal level (calibrated ⇒ observed = nominal). Sharpness is the mean 90% interval width; PIT dev. is the KS distance of the PIT to Uniform$(0,1)$ (0 = perfectly calibrated shape). As is honest to report, the intervals are predominantly *over-conservative*: observed coverage exceeds nominal at 83% of dataset/level cells (mean excess +.09), so bands are safe but wider than ideal. The only exceptions are a few marginal undershoots (worst .02) at the tightest levels. Tightening calibration is left to future work. CRPS / sharpness / PIT dev. / MAE: lower ↓ is better.

		Observed coverage						
Dataset	n	@50 (.50)	@80 (.80)	@90 (.90)	@95 (.95)	CRPS↓	Sharp.	PIT dev.↓
Beijing	118,723	0.88	0.97	0.98	0.99	0.134	1.294	0.221
ETTh1	6,306	0.59	0.83	0.89	0.93	0.163	0.931	0.073
Synthetic	9,051	0.64	0.90	0.96	0.98	0.212	1.648	0.073

Table 17: **Wall-clock on a fixed real task** (BeijingAir, first 3000 rows ×132 sensors, 10% MCAR; MAE on standardised data). Single machine, *one CPU core* with BLAS pinned to 2 threads for every method; *no GPU*. Median of 3 runs. CAFÉ matches the references' accuracy at comparable CPU cost while being the only strictly *causal* (point-in-time) low-error method. Deep neural imputers (SAITS/BRITS/Transformer/CSDI) are omitted here as they require GPU *training*; see text for that $\sim 10^3\times$ compute gap.

Method	Backbone	Time (s) ↓	MAE ↓	Causal?	HW
CAFÉ (ours)	factor	**1.16**	**0.134**	✓	**CPU**
SoftImpute	low-rank	1.03	0.513	✗	CPU
TRMF	MF+AR	1.73	0.288	✗	CPU
Linear interp	1D interp	0.01	0.155	✗	CPU
kNN	kNN	1.20	0.192	✗	CPU

B What CAFÉ yields from one causal pass

Every panel below is produced by the *same* forward run of the model on real or synthetic data and read straight from its internal state—no quantity is illustrative. Together they show that a single causal estimator delivers a portfolio of capabilities that normally require separate tools.

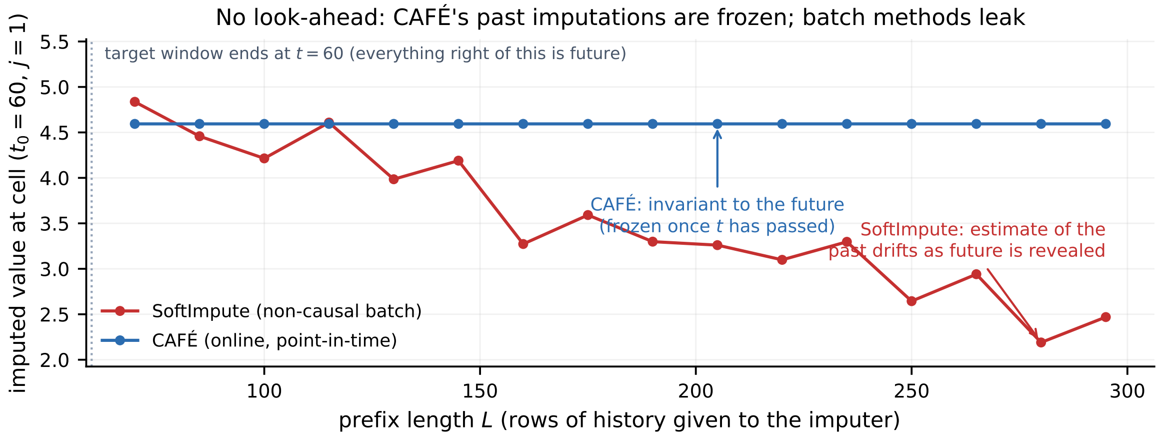


Figure 3: **The moat: no look-ahead.** For a fixed early missing cell, we re-impute on growing time-prefixes. CAFÉ's estimate is *frozen* the moment its time passes (flat line), so a backtest cannot be contaminated; the non-causal batch method's "past" estimate keeps changing as the future is revealed. This is what makes CAFÉ safe for sequential decision-making.

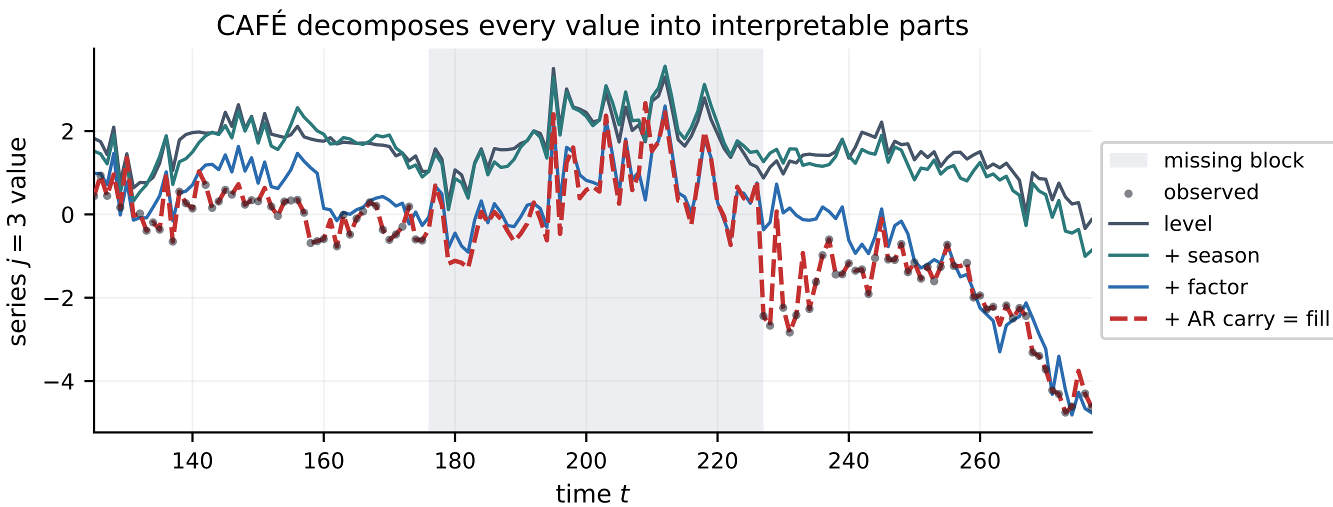
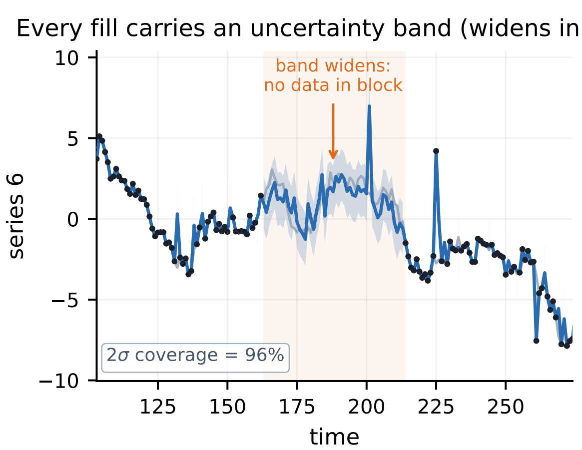


Figure 4: **Interpretability.** The imputed value is the sum of human-readable parts—level, season, shared factors, idiosyncratic carry—read from the trace, not a black box. Each term can be inspected, audited, or used on its own.



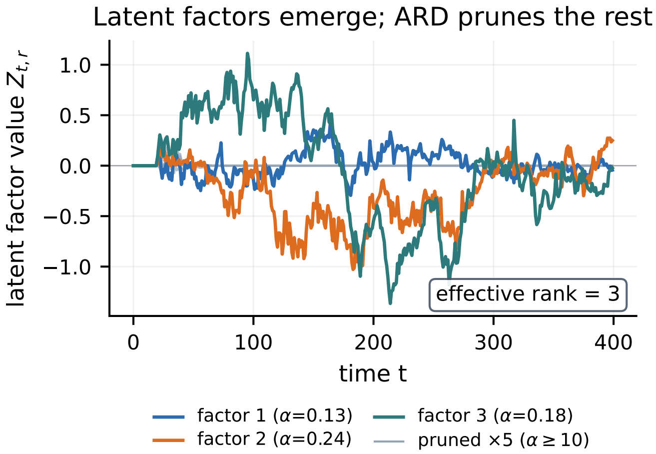


Figure 6: **Latent factors & learned rank.** The common factors z_t (a streaming robust DFM); ARD prunes the rest, so the effective rank is discovered, not set.

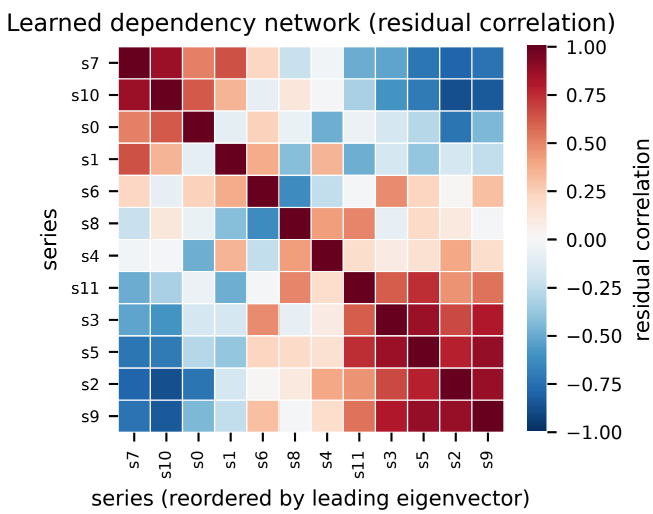


Figure 7: **Dependency network.** The residual covariance recovers the cross-sectional correlation structure between series—useful as features and for contagion analysis.

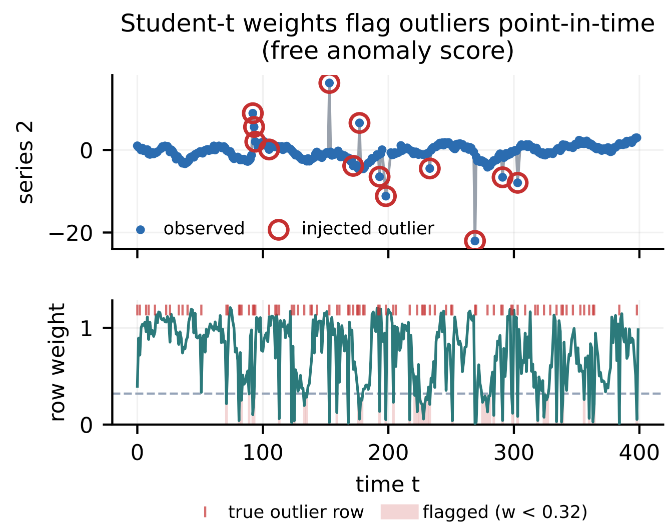


Figure 8: **Anomaly detection (free).** The Student-t weights drop exactly on injected outliers—a point-in-time data-quality / fault score from the same pass.

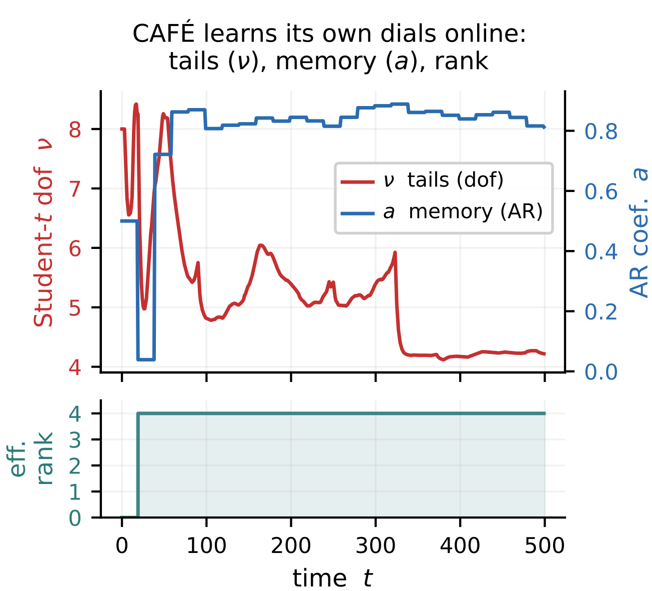


Figure 9: **Self-tuning dials.** The tails (ν), memory (a)

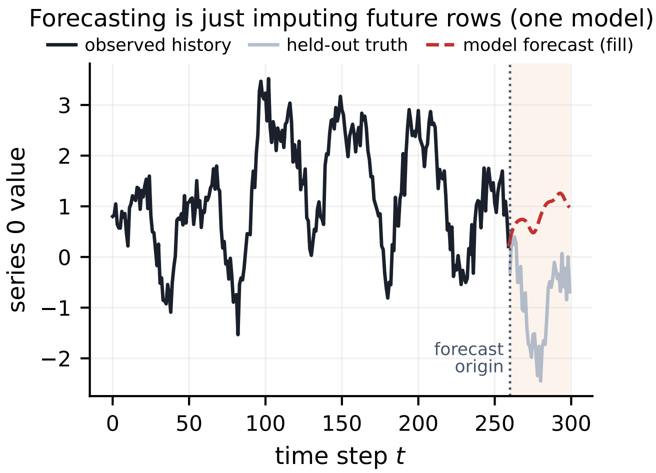


Figure 10: **Forecasting = imputation.** Masking the last rows and imputing them yields a forecast via the AR/Kalman state—one model for fill, forecast and nowcast.

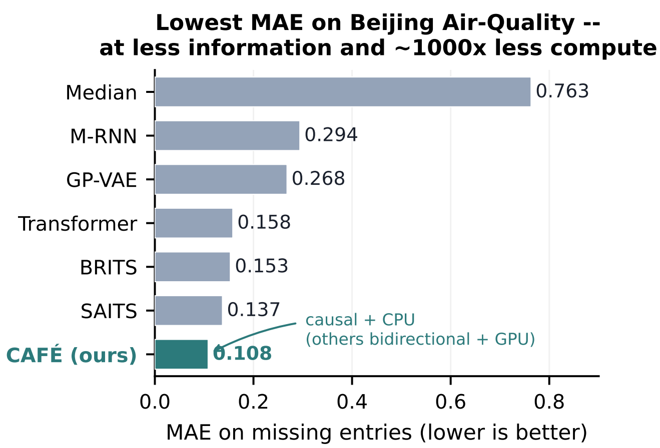


Figure 11: **Accuracy.** CAFÉ's MAE on the named Beijing Air-Quality slice (10% point/MCAR, standardised, 3 seeds) lands in the band of GPU-trained bidirectional deep models—*at far less information and compute, and while remaining causal.* Published deep numbers (different protocols) are context, not a head-to-head ranking; see Table 5.

C Experimental figures

These panels accompany the experiments of §3; every number is from a real run on the stated data.

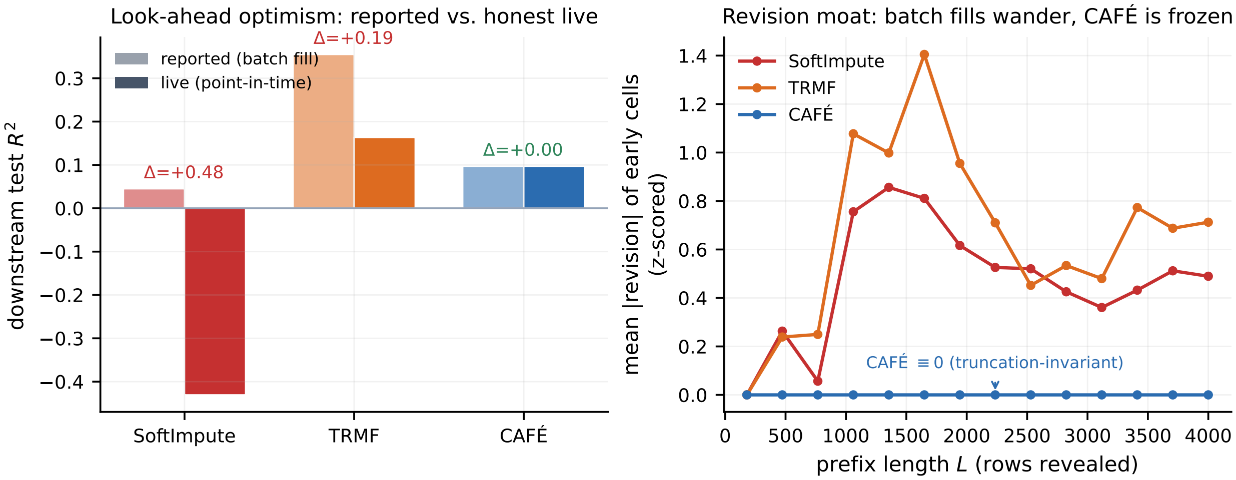


Figure 12: **The cost of look-ahead (Table 8).** *Left:* a downstream ridge forecaster's reported vs. honest *live* R^2 on ETTh1; batch (bidirectional) imputation inflates the reported score, and the same fixed model is worse live, while CAFÉ is identical both ways. *Right:* mean |revision| of an early imputed cell as the future is revealed—batch methods keep rewriting their own past, CAFÉ is flat at zero.

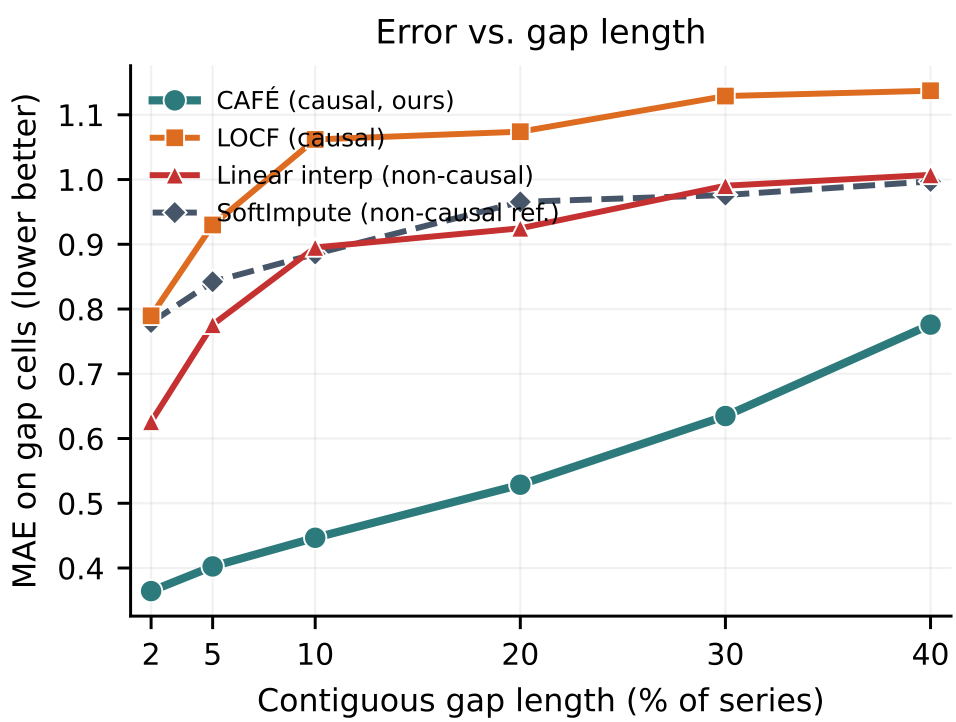


Figure 13: **Graceful degradation under long gaps (Table 14).** Imputation MAE as a single contiguous gap grows to 40% of the series. Local causal methods (LOCF, linear) saturate near the unconditional error; CAFÉ stays bounded and beats even the non-causal SoftImpute reference at every length.

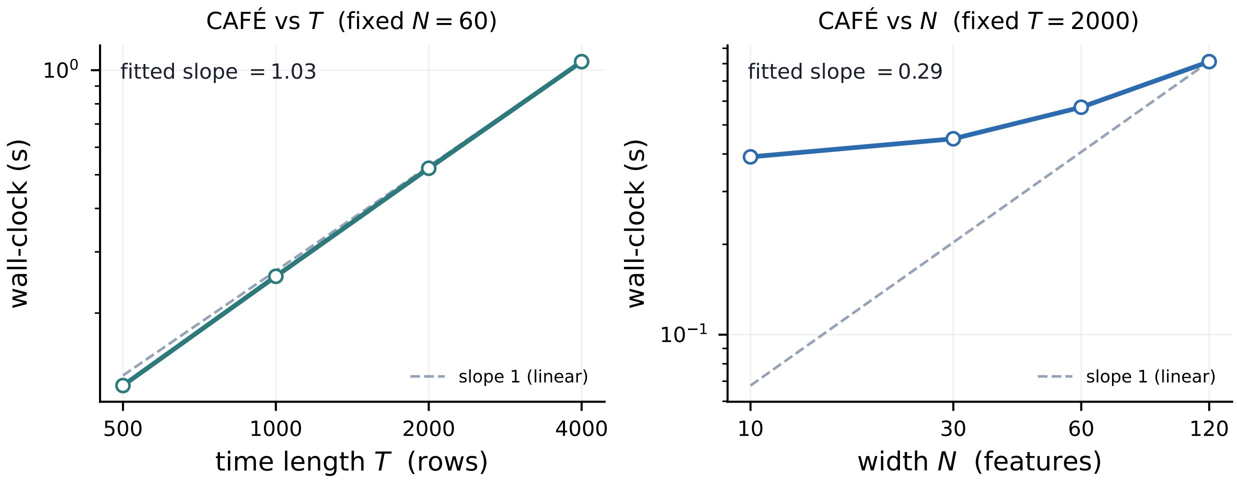


Figure 14: **Scaling (Table 17).** Wall-clock vs. series length T (near-linear, log-log slope ≈ 1) and vs. width N (strongly sub-linear): each row uses only rank-R Cholesky solves, never an $N \times N$ inverse.

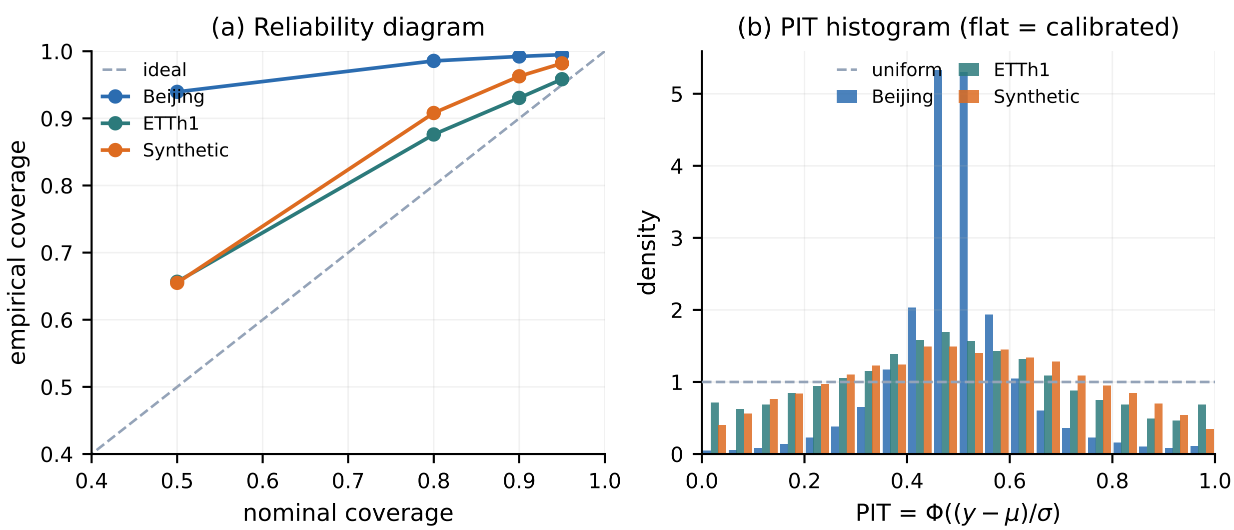


Figure 15: **Calibration (Table 16).** *(a)* Reliability diagram: empirical vs. nominal coverage—curves sit *on or above* the diagonal at most levels (predominantly over-covering), with only marginal undershoots at the tightest levels. *(b)* PIT histograms: a flat profile would be sharply calibrated; CAFÉ's are centre-peaked, the signature of conservative (over-dispersed) intervals—safe, but loose, on the smoother panels.

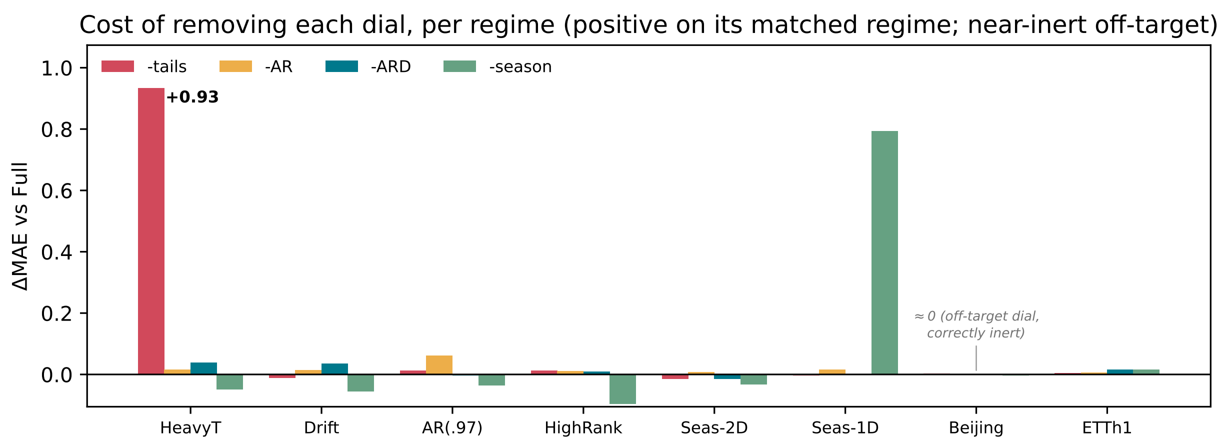


Figure 16: **Full ablation matrix (Table 12).** Cost of removing each dial (ΔMAE vs. full CAFÉ) across all eight regimes. Each dial spikes on the regime it targets and is near-inert—small, often slightly negative—elsewhere, the price of empirical-Bayes priors that shrink an unsupported dial toward off rather than exactly to it.

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